

Venkatesh

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PROFESSIONAL SUMMARY

Senior Data Engineering professional with **14+ years of experience**, including **7+ years in modern data engineering**, specializing in designing cloud-native, scalable data platforms and distributed data processing systems. - Proven expertise in building and optimizing high-volume **ETL/ELT pipelines** (batch & real-time) using **Apache Spark, Airflow, and Kafka**, enabling analytics, BI, and AI-driven decision systems. - Strong foundation in data platform architecture and solution design, with growing alignment toward Solutions Architect responsibilities, including system design, technology selection, and scalable data ecosystem planning. - Hands-on experience in implementing data reliability, quality, and governance frameworks, ensuring high data trust through SLA-driven monitoring, anomaly detection & secure access controls. - Delivered high-impact transformations, including zero-downtime, zero-data-loss migration across big data platforms, and development of robust data layers supporting enterprise-scale analytics and product use cases.

CORE COMPETENCIES

- Data Platform Engineer
- Big Data Architect
- Solutions Architect

WORK EXPERIENCE

NucleusTeg May 2024 - Present

Senior Data Platform Engineer

- Architecting and delivering scalable batch and real-time data pipelines using Apache Spark and Apache Airflow on AWS EMR, processing terabyte-scale datasets.
- Improved distributed data pipeline performance by reducing job execution time from 2-3 hours to 20-30 minutes (**70-85% improvement**) through advanced Spark tuning (partitioning, AQE, shuffle optimization).
- Delivered a **zero-downtime, zero-data-loss migration** of a multi-TB scale big data platform.
- Architected and supported scalable data platforms processing **10+ TB/day workloads**.
- Implemented data reliability and observability frameworks, resulting in a **30-40% reduction in production incidents**.
- Designed optimized analytics-ready data layers using **Spark, Trino, and Iceberg**, improving query performance by 40-60%.
- Optimized AWS-based infrastructure, achieving **25-35% reduction in cloud compute costs**.

Alation May 2023 - Feb 2024

Staff Technical Engineer

- Designed and developed scalable data pipelines for an enterprise-grade Data Catalog platform.
- Built metadata ingestion frameworks to integrate diverse enterprise data sources.
- Developed advanced data transformation and enrichment logic for lineage tracking and metadata standardization.
- Translated business requirements into scalable data models and analytics-ready schemas.

Acceldata Technology Pvt. Ltd. Apr 2021 - May 2023

Solutions Engineer

- Contributed to the development of a data reliability and observability platform.
- Designed automated data quality frameworks (freshness validation, schema drift detection, volume anomaly checks).
- Developed monitoring and alerting mechanisms to detect delayed or missing data.
- Led root cause analysis and incident resolution efforts to reduce MTTR.

IBM Apr 2017 - Feb 2021

Technology Services Specialist

- Managed and supported distributed data environments, including cluster configuration and performance tuning in the Hadoop ecosystem.
- Developed and maintained ETL pipelines for enterprise reporting and analytics.
- Collaborated with cross-functional teams to troubleshoot production issues and ensure SLA adherence.

Future Focus Pvt. Ltd. (Deployed at TCS)

Sep 2015 - Feb 2017

IT Analyst

- Supported enterprise IT infrastructure and resolved technical issues within SLAs.

TCS

Feb 2015 - Sep 2015

IT Analyst

- Contributed to data analysis and reporting tasks for international clients.

Genpact

Jun 2011 - Feb 2015

Senior Associate

- Handled technical support and data processing for enterprise financial systems.

PROJECTS

EDUCATION

B.Sc. in Computer Science from Sri Venkateswara University, India

2008

CERTIFICATIONS

AWS Certified Solutions Architect - Associate

AWS Certified Machine Learning - Specialty

TECHNICAL SKILLS

Big Data & Distributed Systems: Hadoop (HDFS, YARN, Hive, HBase, Impala, ZooKeeper), Apache Spark (Spark SQL, AQE, Shuffle Optimization), Apache Iceberg, Trino, Presto

Streaming & Event Processing: Apache Kafka

Cloud Platforms: AWS (EMR, S3, EC2, IAM), Microsoft Azure (HDInsight), Google Cloud Platform (Dataproc, BigQuery)

Data Governance & Security: RBAC, Apache Ranger, Kerberos, Data Quality Frameworks, Monitoring, Observability

Databases: PostgreSQL (EDB), Oracle, MySQL, Hive Metastore

Programming & Scripting: Python, SQL, Scala, PySpark, Unix Shell Scripting

Orchestration: Apache Airflow

DevOps & Tools: Git, CI/CD (Jenkins/GitHub Actions exposure), Docker

Operating Systems: Linux, Windows